

Chapter 1: Sets

Class 11 Mathematics – NCERT

1. Introduction and Definitions

The concept of a set is fundamental to modern mathematics and is used in relations, functions, probability, and geometry. The theory was developed by the German mathematician Georg Cantor [1].

What is a Set?

A set is a **well-defined collection of objects** [2].

Understanding "Well-defined":

"Well-defined" means we can definitely decide whether a particular object belongs to the collection or not without any ambiguity. For example, "The rivers of India" is a set because we can list them. However, "The collection of five most renowned mathematicians" is **not** a well-defined set because the criteria for being "renowned" vary from person to person [2].

Notation:

* Sets are usually denoted by capital letters: A, B, C, X, Y, Z [3].

* The elements (members) of a set are represented by small letters: a, b, c, x, y, z [3].

The Greek Symbol Epsilon ($\in$):

If an element a is inside a set A , we say " a belongs to A " and write $a \in A$. If b is not in A , we write $b \notin A$ [3].

- Definition: A set is a well-defined collection of objects [2].
- Symbol: $x \in A$ means x belongs to set A .
- Symbol: $x \notin A$ means x does not belong to set A .

2. Methods of Representing a Set

There are two main ways to represent a set [4]:

1. Roster or Tabular Form:

In this form, all elements are listed, separated by commas, and enclosed within braces $\{\}$.

* **Order does not matter:** The set $\{1, 2, 3\}$ is the same as $\{2, 3, 1\}$ [5].

* **No Repetition:** Elements are generally not repeated. For example, the set of letters in "SCHOOL" is $\{S, C, H, O, L\}$ [5, 6].

2. Set-Builder Form:

Instead of listing elements, we write the single common property possessed by all elements. We use a variable (usually x) followed by a colon ":" (which means "such that") and then the property [6, 7].

Format: $V = \{x : x \text{ is a vowel in English alphabet}\}$

Read as: "The set of all x such that x is a vowel in the English alphabet" [7].

- Roster Form: List elements, e.g., $\{2, 4, 6\}$. Order is immaterial [4].
- Set-Builder Form: Describe property, e.g., $\{x : x \text{ is a natural number less than } 4\}$ [7].
- Tip: In Roster form, do not repeat elements (e.g., 'SCHOOL' becomes S, C, H, O, L) [6].

3. Solved Textbook Examples (Step-by-Step)

Example 1: Write the solution set of $x^2 + x - 2 = 0$ in roster form.

Solution: Solve the quadratic equation:

$$(x - 1)(x + 2) = 0 \implies x = 1, -2.$$

Therefore, the set is $\{1, -2\}$ [8].

Example 2: Write the set $\{x : x \text{ is a positive integer and } x^2 < 40\}$ in roster form.

Solution: We check squares of positive integers:

$$1^2 = 1 < 40$$

$$2^2 = 4 < 40$$

$$3^2 = 9 < 40$$

$$4^2 = 16 < 40$$

$$5^2 = 25 < 40$$

$$6^2 = 36 < 40$$

$$7^2 = 49 > 40 \text{ (Stop here).}$$

The elements are 1, 2, 3, 4, 5, 6. The set is $\{1, 2, 3, 4, 5, 6\}$ [8].

Example 3: Write $A = \{1, 4, 9, 16, 25, \dots\}$ in set-builder form.

Solution: Observe the pattern. These are squares of natural numbers ($1^2, 2^2, 3^2, \dots$).

$$A = \{x : x = n^2, \text{ where } n \in N\} \text{ [9].}$$

Example 5: Match the sets.

(i) $\{P, R, I, N, C, A, L\}$ matches with $\{x : x \text{ is a letter of the word PRINCIPAL}\}$ (Note the removal of duplicate letters P and I) [10, 11].

(ii) $\{0\}$ matches with $\{x : x \text{ is an integer and } x + 1 = 1\}$ because $x + 1 = 1 \implies x = 0$ [10, 11].

- Always solve the equation or condition first before listing elements for Roster form.
- Look for the mathematical pattern (squares, odd numbers, divisors) for Set-Builder form.

4. Detailed Solutions: Exercise 1.1

Below are the comprehensive solutions for NCERT Exercise 1.1 [11-15].

Q1. Which of the following are sets? Justify.

*** (i) The collection of all the months of a year beginning with the letter J.**

*** Solution:** It is a **Set**. The collection is well-defined (January, June, July).

*** (ii) The collection of ten most talented writers of India.**

*** Solution:** **Not a Set**. "Most talented" is vague and varies from person to person [2].

*** (iii) A team of eleven best-cricket batsmen of the world.**

*** Solution:** **Not a Set**. "Best" is subjective and not well-defined.

*** (iv) The collection of all boys in your class.**

* **Solution:** It is a **Set**. It is well-defined; we can identify exactly who belongs to this group.

* **(v) The collection of all natural numbers less than 100.**

* **Solution:** It is a **Set**. Elements are fixed (1 to 99).

* **(vi) A collection of novels written by Munshi Prem Chand.**

* **Solution:** It is a **Set**. The books written by him are definite and known.

* **(vii) The collection of all even integers.**

* **Solution:** It is a **Set** (Infinite, but well-defined).

* **(viii) The collection of questions in this Chapter.**

* **Solution:** It is a **Set**.

* **(ix) A collection of most dangerous animals of the world.**

* **Solution:** **Not a Set**. "Most dangerous" is subjective.

Q2. Let $A = \{1, 2, 3, 4, 5, 6\}$. Insert appropriate symbol $\in$ or $\notin$.

* (i) $5 \in A$ (5 is in the list)

* (ii) $8 \notin A$ (8 is not in the list)

* (iii) $0 \notin A$

* (iv) $4 \in A$

* (v) $2 \in A$

* (vi) $10 \notin A$ [12].

Q3. Write in Roster Form:

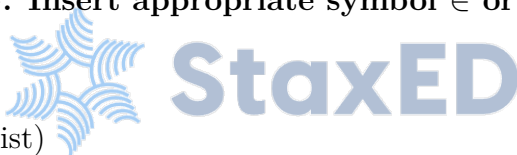
* **(i) $A = \{x : x \text{ is an integer and } -3 \leq x < 7\}$**

* **Solution:** Integers between -3 (inclusive) and 7 (exclusive). $A = \{-3, -2, -1, 0, 1, 2, 3, 4, 5, 6\}$.

* **(ii) $B = \{x : x \text{ is a natural number less than 6}\}$**

* **Solution:** Natural numbers start from 1. $B = \{1, 2, 3, 4, 5\}$.

* **(iii) $C = \{x : x \text{ is a two-digit natural number such that sum of its digits is 8}\}$**



* **Solution:** Pairs adding to 8: (1,7), (2,6), (3,5), (4,4), (5,3), (6,2), (7,1), (8,0). $C = \{17, 26, 35, 44, 53, 62, 71, 80\}$.

* **(iv) $D = \{x : x \text{ is a prime number which is divisor of } 60\}$**

* **Solution:** Divisors of 60: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60. Prime ones are 2, 3, 5.
 $D = \{2, 3, 5\}$.

* **(v) $E = \text{The set of all letters in the word TRIGONOMETRY}$**

* **Solution:** List letters without repetition. $E = \{T, R, I, G, O, N, M, E, Y\}$.

* **(vi) $F = \text{The set of all letters in the word BETTER}$**

* **Solution:** $F = \{B, E, T, R\}$.

Q4. Write in Set-Builder Form:

* **(i) $\{3, 6, 9, 12\}$**

* **Solution:** Multiples of 3. $\{x : x = 3n, n \in N \text{ and } 1 \leq n \leq 4\}$.

* **(ii) $\{2, 4, 8, 16, 32\}$**

* **Solution:** Powers of 2 (2^1 to 2^5). $\{x : x = 2^n, n \in N \text{ and } 1 \leq n \leq 5\}$.

* **(iii) $\{5, 25, 125, 625\}$**

* **Solution:** Powers of 5 (5^1 to 5^4). $\{x : x = 5^n, n \in N \text{ and } 1 \leq n \leq 4\}$.

* **(iv) $\{2, 4, 6, \dots\}$**

* **Solution:** Infinite set of even natural numbers. $\{x : x \text{ is an even natural number}\}$.

* **(v) $\{1, 4, 9, \dots, 100\}$**

* **Solution:** Perfect squares up to 10^2 . $\{x : x = n^2, n \in N \text{ and } 1 \leq n \leq 10\}$.

Q5. List all elements:

* **(i) Odd natural numbers: $\{1, 3, 5, \dots\}$.**

* **(ii) Integer $-\frac{1}{2} < x < \frac{9}{2}$: $-0.5 < x < 4.5$. Integers are $\{0, 1, 2, 3, 4\}$.**

* **(iii) Integer $x^2 \leq 4$:** Squares less than or equal to 4 are 0, 1, 4. Roots are -2, -1, 0, 1, 2.
Set: $\{-2, -1, 0, 1, 2\}$.

* **(iv) Letter in "LOYAL": $\{L, O, Y, A\}$ (Don't repeat L).**

- * (v) **Month not having 31 days:** $\{February, April, June, September, November\}$.
- * (vi) **Consonant preceding k:** $\{b, c, d, f, g, h, j\}$ (Skip vowels a, e, i).

Q6. Match the sets:

- * (i) $\{1, 2, 3, 6\}$ matches (c) Natural number and divisor of 6.
 - * (ii) $\{2, 3\}$ matches (a) Prime number and divisor of 6.
 - * (iii) $\{M, A, T, H, E, I, C, S\}$ matches (d) Letters of MATHEMATICS.
 - * (iv) $\{1, 3, 5, 7, 9\}$ matches (b) Odd natural number less than 10 [14, 15].
- Mistake Alert (Q1): Subjective terms like "best", "talented", or "dangerous" never form well-defined sets.
 - Mistake Alert (Q3/Q5): When listing letters of a word, never repeat letters (e.g., BETTER becomes B, E, T, R).
 - Tip (Q4): Identify the pattern $(n, n^2, 2^n)$ and define the range of n clearly.

5. Types of Sets: Empty, Finite, Infinite, and Equal

We classify sets based on the number of elements they contain and their relationship with other sets.

1. The Empty Set:

A set which does not contain any element is called the empty set, null set, or void set. It is denoted by the symbol ϕ or $\{\}$.

Example: $A = \{x : 1 < x < 2, x \text{ is a natural number}\}$. Since there is no natural number between 1 and 2, A is an empty set.

2. Finite and Infinite Sets:

* **Finite Set:** A set which is empty or consists of a definite number of elements. The number of distinct elements in a finite set A is denoted by $n(A)$.

* **Infinite Set:** A set whose elements cannot be counted or listed completely because they are infinite in number. For example, the set of natural numbers N .

3. Equal Sets:

Two sets A and B are said to be equal if they have exactly the same elements. We write $A = B$. If they are not equal, $A \neq B$.

Note: The order of elements or repetition of elements does not change the set. $\{1, 2, 3\} = \{3, 1, 2\} = \{1, 1, 2, 3\}$.

- Empty Set (ϕ): Contains no elements [1].
- Finite Set: Countable number of elements [2].
- Infinite Set: Elements continue indefinitely (e.g., points on a line) [2].
- Equal Sets: A and B have exactly the same elements [3].

6. Detailed Solutions: Exercise 1.2

Q1. Which are null sets?

- * (i) Set of odd natural numbers divisible by 2: **Null Set** (No odd number is divisible by 2).
- * (ii) Set of even prime numbers: **Not Null** (Contains the element 2).
- * (iii) $\{x : x < 5 \text{ and } x > 7\}$: **Null Set** (A number cannot be less than 5 and greater than 7 simultaneously).
- * (iv) Point common to two parallel lines: **Null Set** (Parallel lines never meet).

Q2. Finite or Infinite?

- * (i) Months of a year: **Finite** (12 elements).
- * (ii) $\{1, 2, 3, \dots\}$: **Infinite** (Natural numbers).
- * (iii) $\{1, 2, \dots, 99, 100\}$: **Finite**.
- * (iv) Positive integers ≤ 100 : **Infinite**.
- * (v) Prime numbers ≤ 99 : **Finite**.

Q3. State Finite or Infinite:

- * (i) Lines parallel to x-axis: **Infinite**.
- * (ii) Letters in English alphabet: **Finite** (26).
- * (iii) Multiples of 5: **Infinite**.

* (iv) Animals living on earth: **Finite** (Though large, the number is definite).

* (v) Circles passing through origin (0,0): **Infinite**.

Q4. State whether $A = B$:

* (i) $A = \{a, b, c, d\}, B = \{d, c, b, a\}$: **Yes, $A=B$.**

* (ii) $A = \{4, 8, 12, 16\}, B = \{8, 4, 16, 18\}$: **No** ($12 \in A$ but $12 \notin B$).

* (iii) $A = \{2, 4, 6, 8, 10\}, B = \{x : x \text{ is positive even integer } \leq 10\}$: **Yes, $A=B$.**

* (iv) $A = \{x : x \text{ is multiple of } 10\}, B = \{10, 15, 20, \dots\}$: **No** ($15 \in B$ but not in A).

- Tip: To check equality, verify every element of A is in B and every element of B is in A .
- Observation: Infinite sets are often represented by three dots (...) [4].

7. Subsets and Intervals

Subsets:

A set A is a subset of set B if every element of A is also an element of B . We write $A \subset B$.

* Symbol: $A \subset B$ if $a \in A \Rightarrow a \in B$.

* **Proper Subset:** If $A \subset B$ and $A \neq B$, then A is a proper subset of B .

* **Singleton Set:** A set with only one element, e.g., $\{a\}$.

* **Universal Set (U):** A basic set containing all sets under consideration. For example, in number theory, the set of Real Numbers (R) can be the universal set.

Intervals as Subsets of R :

Let $a, b \in R$ and $a < b$.

1. **Open Interval (a, b) :** $\{x : a < x < b\}$ (End points excluded).

2. **Closed Interval $[a, b]$:** $\{x : a \leq x \leq b\}$ (End points included).

3. **Semi-open Intervals:**

* $[a, b) = \{x : a \leq x < b\}$ (Includes a , excludes b).

* $(a, b] = \{x : a < x \leq b\}$ (Excludes a , includes b).

- Every set is a subset of itself ($A \subset A$) [5].
- The empty set ϕ is a subset of every set [5].
- Mathematical Logic: $A \subset B \iff (x \in A \Rightarrow x \in B)$ [6].
- Length of interval: The length of any interval (a, b) or $[a, b]$ is $b - a$ [7].

8. Detailed Solutions: Exercise 1.3

Q1. Fill with $\subset$ or $\not\subset$:

- * (i) $\{2, 3, 4\} \subset \{1, 2, 3, 4, 5\}$
- * (ii) $\{a, b, c\} \not\subset \{b, c, d\}$ (a is missing in the second set).
- * (iii) Student of Class XI $\subset$ Student of your school.
- * (iv) Circle in plane $\not\subset$ Circle with radius 1 (The first set is larger).
- * (v) Triangle in plane $\not\subset$ Rectangle in plane.
- * (vi) Equilateral triangle $\subset$ Triangle in same plane.
- * (vii) Even natural number $\subset$ Integer.

Q4. Write down all subsets:

- * (i) $\{a\}$: Subsets are $\phi, \{a\}$.
- * (ii) $\{a, b\}$: Subsets are $\phi, \{a\}, \{b\}, \{a, b\}$.
- * (iii) $\{1, 2, 3\}$: Subsets are $\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$.
- * (iv) ϕ : Subset is ϕ .

Q6. Write as intervals:

- * (i) $\{x : -4 < x \leq 6\} \Rightarrow (-4, 6]$.
- * (ii) $\{x : -12 < x < -10\} \Rightarrow (-12, -10)$.
- * (iii) $\{x : 0 \leq x < 7\} \Rightarrow [0, 7)$.
- * (iv) $\{x : 3 \leq x \leq 4\} \Rightarrow [3, 4]$.

Q7. Universal Set Proposal:

- * (i) Set of right triangles: $U = \text{Set of all triangles}$.
- * (ii) Set of isosceles triangles: $U = \text{Set of all triangles}$.
- Important: The element of a set can be a set itself. If $A = \{1, \{2, 3\}\}$, then $\{2, 3\} \in A$, but $\{2, 3\} \subset A$ is false. $\{\{2, 3\}\} \subset A$ is true [10].
- Total subsets: A set with n elements has 2^n subsets (e.g., $\{1, 2, 3\}$ has $2^3 = 8$ subsets) [11].

9. Venn Diagrams and Operations on Sets

Venn Diagrams:

Geometric representations where the Universal Set (U) is a rectangle and subsets are circles inside it.

1. Union of Sets ($A \cup B$):

The set of elements which are in A **or** in B (or in both).

* $A \cup B = \{x : x \in A \text{ or } x \in B\}$.

* *Properties:* $A \cup B = B \cup A$ (Commutative), $A \cup \phi = A$.

2. Intersection of Sets ($A \cap B$):

The set of elements which are common to both A **and** B .

* $A \cap B = \{x : x \in A \text{ and } x \in B\}$.

* **Disjoint Sets:** If $A \cap B = \phi$ (no common elements), the sets are disjoint.

* *Properties:* $A \cap B = B \cap A$, $A \cap \phi = \phi$.

* *Distributive Law:* $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

3. Difference of Sets ($A - B$):

The set of elements belonging to A but **not** to B .

* $A - B = \{x : x \in A \text{ and } x \notin B\}$.

* *Note:* In general, $A - B \neq B - A$.

- Union ($A \cup B$): Combine all elements, take common ones only once [12].

- Intersection ($A \cap B$): Take only common elements [13].
- Difference ($A - B$): Remove elements of B from A [14].

10. Detailed Solutions: Exercise 1.4

Q1. Find the Union:

- * (i) $X = \{1, 3, 5\}, Y = \{1, 2, 3\} \Rightarrow X \cup Y = \{1, 2, 3, 5\}$.
- * (v) $A = \{1, 2, 3\}, B = \phi \Rightarrow A \cup B = \{1, 2, 3\}$.

Q5. Find the Intersection (from Q1 sets):

- * (i) $X \cap Y = \{1, 3\}$ (Common elements).
- * (ii) $A = \{a, e, i, o, u\}, B = \{a, b, c\} \Rightarrow A \cap B = \{a\}$.

Q9. Difference of Sets:

Given $A = \{3, 6, 9, 12, 15, 18, 21\}, B = \{4, 8, 12, 16, 20\}, C = \{2, 4, 6, 8, 10, 12, 14, 16\}, D = \{5, 10, 15, 20\}$.

- * (i) $A - B$: Elements in A excluding 12. $\{3, 6, 9, 15, 18, 21\}$.
- * (ii) $A - C$: Elements in A excluding 6, 12. $\{3, 9, 15, 18, 21\}$.
- * (iii) $B - A$: Elements in B excluding 12. $\{4, 8, 16, 20\}$.

Q12. True or False (Disjoint Sets):

- * (i) $\{2, 3, 4, 5\}$ and $\{3, 6\}$: **False** (3 is common).
- * (ii) $\{a, e, i, o, u\}$ and $\{a, b, c, d\}$: **False** (a is common).
- * (iii) $\{2, 6, 10, 14\}$ and $\{3, 7, 11, 15\}$: **True** (No common elements).

- Common Mistake: In $A - B$, do not include elements of B that are not in A. Only focus on what remains in A.
- Disjoint means intersection is ϕ [15].

11. Complement of a Set

Definition:

Let U be the universal set and A be a subset of U . The complement of A (denoted A') is the set of all elements of U which are not in A .

* $A' = \{x : x \in U \text{ and } x \notin A\}.$

* Mathematically: $A' = U - A.$

De Morgan's Laws:

1. $(A \cup B)' = A' \cap B'$ (The complement of the union is the intersection of complements).

2. $(A \cap B)' = A' \cup B'$ (The complement of the intersection is the union of complements).

Other Properties:

* $(A')' = A$ (Law of double complementation).

* $A \cup A' = U$ and $A \cap A' = \phi.$

* $\phi' = U$ and $U' = \phi.$

• Formula: $A' = U - A$ [16].

• De Morgan's Law 1: $(A \cup B)' = A' \cap B'$ [17].

• De Morgan's Law 2: $(A \cap B)' = A' \cup B'$ [17].

12. Detailed Solutions: Exercise 1.5

Q1. Given $U = \{1...9\}, A = \{1, 2, 3, 4\}, B = \{2, 4, 6, 8\}, C = \{3, 4, 5, 6\}:$

* (i) $A' = U - A = \{5, 6, 7, 8, 9\}.$

* (ii) $B' = U - B = \{1, 3, 5, 7, 9\}.$

* (iii) $(A \cup C)'$: First $A \cup C = \{1, 2, 3, 4, 5, 6\}$. Then complement is $\{7, 8, 9\}.$

* (v) $(A')'$: This returns to $A = \{1, 2, 3, 4\}.$

Q3. Complements (Natural numbers N is Universal):

* (i) $\{x : x \text{ is even}\}' = \{x : x \text{ is odd natural number}\}.$

* (iii) $\{x : x \text{ is multiple of } 3\}' = \{x : x \text{ is not a multiple of } 3\}.$

* (iv) $\{x : x \text{ is prime}\}' = \{x : x \text{ is positive composite number and } 1\}.$

Q5. Venn Diagrams:

* (i) $(A \cup B)'$: Shade the region outside both circles A and B.

* (ii) $A' \cap B'$: Same as above (De Morgan's Law).

Q7. Fill in the blanks:

* (i) $A \cup A' = U$

* (ii) $\phi' \cap A = U \cap A = A$

* (iii) $A \cap A' = \phi$

* (iv) $U' \cap A = \phi \cap A = \phi$

- Key Concept: $A \cup A'$ fills the entire Universe U [18].
- Key Concept: There is no overlap between a set and its complement ($A \cap A' = \phi$).

